

Chetan Gadhiya

(+91) 9925885858 | chetan.certi.001@gmail.com | linkedin/chetangadhiya | github.com/chetangadhiya5062 | chetangadhiya.vercel.app

EDUCATION

Pandit Deendayal Energy University

Jul 2023 - May 2027

Bachelor of Technology in Computer Science Engineering (CGPA: 8.71/10)

EXPERIENCE

Research & Innovation Intern [GitHub]

May 2026 - Jul 2026

HNNOIX - NeuroFlow AI

- Contributed to the development of NeuroFlow AI, a production-grade AI Operating Platform supporting Autonomous AI Agents, RAG, **Knowledge Graphs**, **Workflow Orchestration**, and enterprise AI applications at scale
- Engineered reusable AI infrastructure including a provider-agnostic **LLM Gateway**, **RAG Runtime**, **AI Memory Layer**, **Knowledge Base**, and modular runtimes for agent execution, prompt orchestration, and intelligent knowledge retrieval
- Established a scalable AI platform using Clean Architecture, SOLID principles, interface-first design, structured observability, automated testing, CI workflows, and containerized deployment for reliable and maintainable enterprise AI systems

Freelance Generative AI Engineer

Dec 2024 - Jun 2025

Independent AI Development

- Developed **2 AI systems** using **multi-agent and RAG-based approaches** for automated fact verification and knowledge-grounded question answering, transforming complex requirements into AI-driven solutions
- Built a **CrewAI**-based multi-agent pipeline integrating **Gemini**, **SerperDev**, **Pydantic**, and **FastAPI** for claim extraction, web research, structured verdict generation, confidence scoring, and citation-based verification
- Reduced inference overhead by **15%** through conditional processing and workflow optimization in the fact-checking pipeline, minimizing unnecessary LLM execution while maintaining structured outputs and response reliability
- Developed a **RAG pipeline using AWS Bedrock**, implementing embedding-based retrieval, semantic search, prompt engineering, and LLM-based response generation over structured and unstructured knowledge

SELECTED PROJECTS

OpenEnv Data Cleaning RL Agent

[GitHub]

- Built a custom **Reinforcement Learning (RL)** environment where a **Llama 3** agent autonomously cleans tabular data
- Designed a reward-driven policy with custom reward functions (+2 missing value reduction, +3 deduplication, +5 completion, penalties for redundant actions) and step constraints, including anti-loop penalties for repeated actions
- Implemented **Pydantic** schemas and a prompt-engineered **LLM pipeline** with structured **JSON** outputs, parsing logic, and fallback mechanisms, incorporating state, action, and reward history into agent decisions for iterative data cleaning
- Developed a **FastAPI** server with **REST/WebSocket** support and deployed via multi-stage **Docker** on **HuggingFace Spaces**, exposing real-time environment state and agent actions through API endpoints for agent execution and evaluation

AetherMail - Autonomous AI Email Management System

[GitHub]

- Developed a stateful agentic workflow using **LangGraph** (Gemini 2.5 Flash) enabling multi-step reasoning, tool invocation, and autonomous email actions across **6 specialized tools** for search, organization, drafting, and bulk operations
- Engineered a **RAG** pipeline using **Ollama** embeddings and **Qdrant** for semantic retrieval over unstructured emails
- Built a scalable pipeline (**Kafka** → **Spark** → **HDFS/PostgreSQL/Qdrant**) with delegation-based execution for bulk tasks, processing 768-dimensional embeddings alongside encrypted email storage and metadata persistence

TECHNICAL SKILLS

AI/ML: Machine Learning, Reinforcement Learning, Deep Learning, Natural Language Processing, LLMs, RAG

GenAI & Agentic AI: Multi-Agent Systems, AI Agents, CrewAI, LangGraph, Gemini, Llama 3, AWS Bedrock

AI Engineering: Prompt Engineering, Pydantic, FastAPI, Vector Databases, Qdrant, REST APIs, WebSockets

Data & Distributed Systems: Apache Kafka, Apache Spark, PostgreSQL, MongoDB, MySQL

Programming: Python, SQL **Tools & Deployment:** Docker, Git, GitHub

Core CS: Data Structures & Algorithms, Problem Solving (LeetCode)

ACHIEVEMENTS & LEADERSHIP

- IEEE AIMV 2025 Conference (PDEU):** Lead Student Volunteer managing operations for **160+ research papers**; coordinated a **35-member volunteer team** and presented **two research papers** on behalf of authors
- Hackathons:** Selected among **1200+ teams** in Code4Cause 2.0 (NSUT); **Top 25** in SIH internal rounds for **two consecutive years**; participated in **GenAI Exchange Hackathon** and **HACKOUT'24 (DAIICT)**
- Encode (AI/ML):** PDEU coding club member for **2 years**, contributing to AI/ML peer learning and discussions
- Certifications:** Oracle Cloud Infrastructure Certified Foundations Associate; AWS Generative AI Developer Learning Plan; Deep Learning - NPTEL (IIT Ropar, **73%**); AI-ML for Geodata Analysis - **ISRO-IIRS**